

HAT-P-27b

R-0032

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-27_260422 · created 2026-07-08T20:07:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461152.6966 +/- 0.0089 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.093 +/- 0.039
Transit depth (Rp/Rs)^2	0.86 +/- 0.72 [%]
Orbital Inclination (inc)	84.58 +/- 2.91
Airmass coefficient 1 (a1)	1.22 +/- 0.65
Airmass coefficient 2 (a2)	-0.12 +/- 0.25
Scatter in the residuals of the lightcurve fit is	54.63 %
Best Comparison Star	None
Optimal Aperture	1.04
Optimal Annulus	3.15
Transit Duration (day)	0.038 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.093 ± 0.039 vs 0.119 (deviation 0.67σ)
Duration fit vs published	0.038 d vs 0.07 d (deviation 1.14σ)
Mid-time O–C	-21.4 min
Depth SNR	2.4
Residual scatter	54.63 %

Light curve

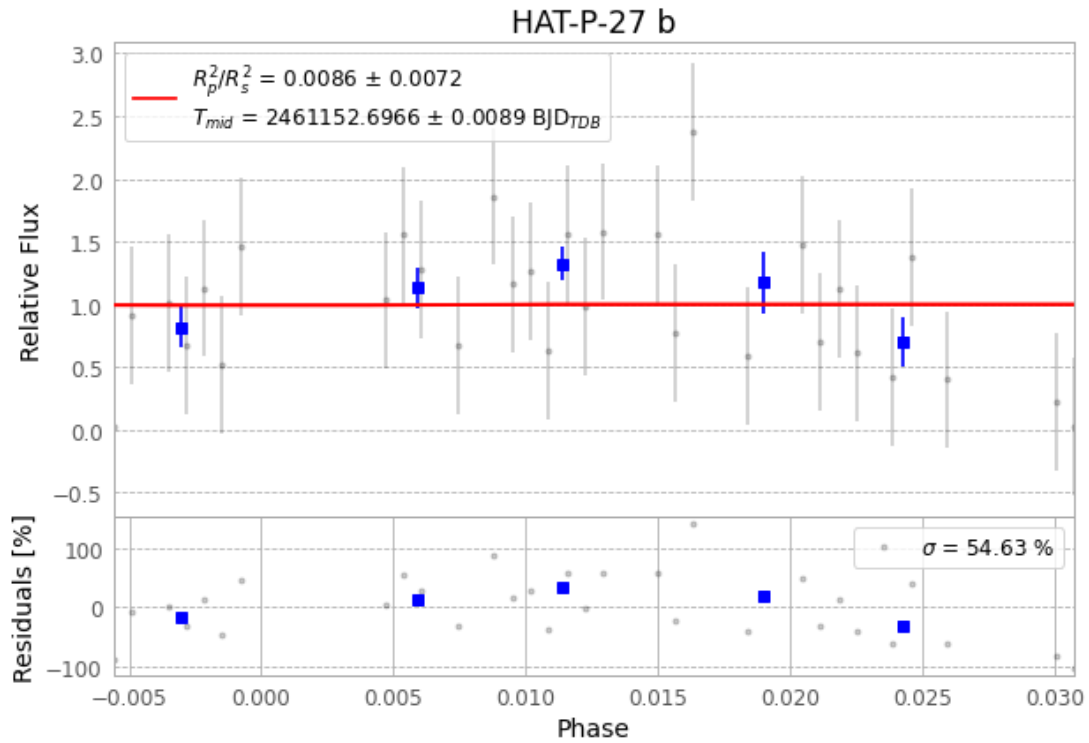


Figure 1 — FinalLightCurve_HAT-P-27 b_2026-04-21.png (SHA-256 5ab199b840d2760b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [137, 417], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b4507aa492151fe8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

54 FITS frames (35 MB), HATP-27260422040916.FITS → HATP-27260422064816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-27-260422.log (604cae224a50...), FinalLightCurve_HAT-P-27 b_2026-04-21.png (5ab199b840d2...), FinalLightCurve_HAT-P-27 b_2026-04-21.csv (d093ca5d6715...)

Compute resources

Wall time 164.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.